

Q1: NOV 2001 P2

- 1 A water-wheel has eight buckets equally spaced around its circumference, as illustrated in Fig. 1.1.

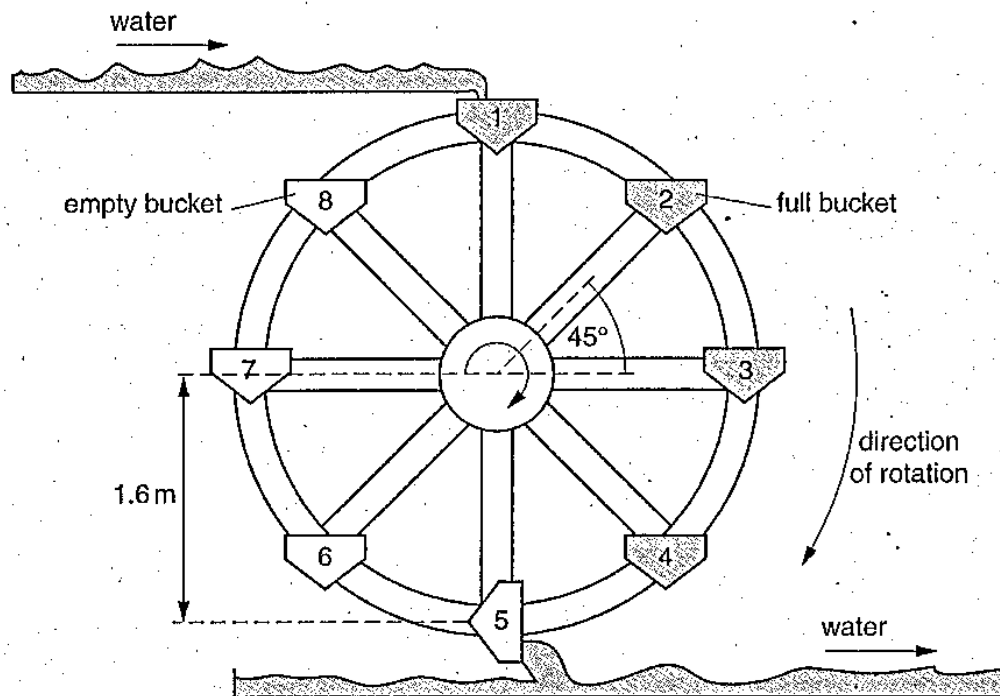


Fig. 1.1

The distance between the centre of each bucket and the centre of the wheel is 1.6 m. When a bucket is at its highest point, the bucket is filled with a mass of 40 kg of water. The wheel rotates and the bucket is emptied at its lowest point.

- (a) (i) Define the *moment of a force*.

.....

.....

- (ii) Write down the number of the bucket that provides the largest moment about the axle of the wheel.

.....

- (iii) Write down the numbers of those buckets containing water that cause a moment about the axle.

.....

- (iv) Calculate, for the wheel in the position shown in Fig. 1.1, the total resultant moment about the centre of the wheel of the water in the buckets.

Q2:NOV 2002 P2

- 3 A 'torque wrench' is a tool used to tighten nuts and bolts to a precise measurement. One model is shown in Fig. 3.1.

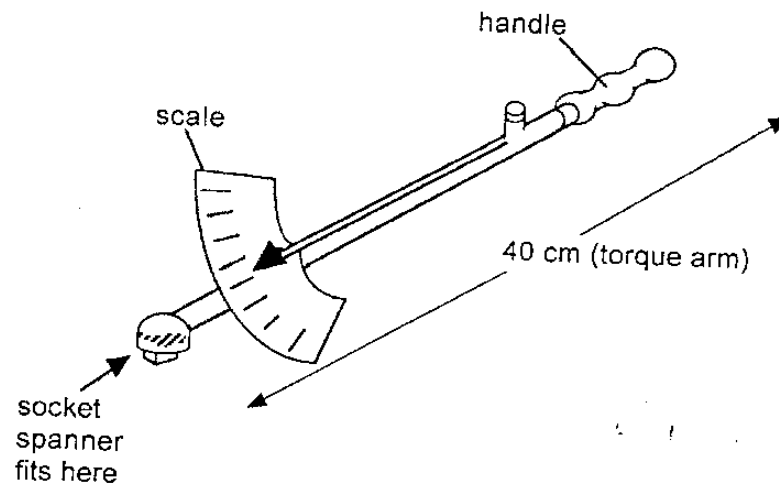


Fig. 3.1

- (a) (i) Explain what is meant by a *torque*.

[2]

- (ii) Explain briefly how the torque wrench is used to tighten a nut by the required amount.

[2]

- (iii) Calculate the torque for the wrench shown if the force applied to the end of the handle is 200 N.

torque = _____

[2]

- (b) (i) Why does a torque wrench have a long handle or torque arm?

[2]

2.3 FORCES P2&P3 QUESTIONS

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- (ii) State one risk of using a wrench whose torque arm is too long.

[1]

- (c) (i) Calculate the distance moved by the end tip of the handle during one complete turn when the force in a(iii) is applied.

distance = _____

[2]

Q3:JUN 2008 P2

- 2 A parachutist of mass 55 kg in a free fall dive reached a terminal velocity, v , of 90 ms^{-1} before the parachute opened.

- (a) Define the term *terminal velocity*.

[2]

- (b) Calculate

- (i) the force of air resistance on the diver at terminal velocity,

air resistance = _____

[2]

- (ii) the value of the constant α if

$$\text{air resistance} = \alpha v^2.$$

$$\alpha = \underline{\hspace{2cm}}$$

[2]

Q4:NOV 2013 P2

- 2 (a) State the conditions necessary for a body to be in equilibrium.

[2]

- (b) Fig. 2.1 shows a sphere of mass 2 kg kept in position by two strings, AB and CD. AB is horizontal and CD is inclined at 30° to the vertical.

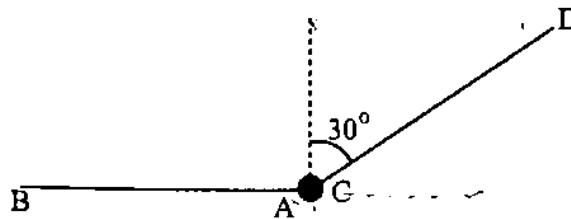


Fig. 2.1

Calculate the tension in each string.

Q5:NOV 2014 P2

- 2 (a) State the **two** conditions for equilibrium.

[2]

- (b) Fig. 2.1 shows a builder's arm holding a brick.

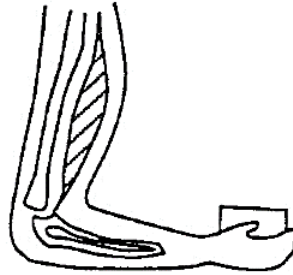


Fig. 2.1

On Fig. 2.1, use arrows to show the points and lines of action of the forces which keep the arm in equilibrium.

[4]

Q6:NOV 2015 P2

- 3 (i) Explain the origin of the upthrust acting on a body in a fluid.

[2]

- (ii) Fig.3.1 shows a steel ball of diameter 8.2 mm falling through oil at a steady speed.

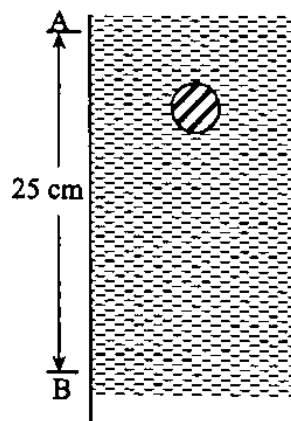


Fig. 3.1

2.3 FORCES P2&P3 QUESTIONS

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The ball moves from A to B in 0.60 s.

Assuming the density of steel and oil to be $7\,800\text{ kgm}^{-3}$ and 900 kgm^{-3} respectively, calculate the

1. weight of the ball,

weight _____

2. upthrust on the ball,

upthrust _____

3. terminal velocity of the ball.

terminal velocity _____

[5]

Q7:JUN 2018 P2

- 3 (a) State the origin of *upthrust*.

[1]

2.3 FORCES P2&P3 QUESTIONS

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- (b) A spherical steel ball of diameter 9.00 mm and density 7.8 gcm^{-3} falls through oil of density 9.0 gcm^{-3} at a steady speed.

(i) Identify a force other than upthrust, that is acting on the ball.

(ii) Calculate the upthrust on the ball.

upthrust = _____ [3]

- (c) (i) State two conditions for a system of forces to be in equilibrium.

1. _____

2. _____

- (ii) Fig. 3.1 shows a ladder of length 12.0 m and weight 500 N propped up against a smooth vertical wall. The lower end is on rough horizontal ground at a distance of 4.0 m from the wall. The upper end is 11.3 m above ground level.

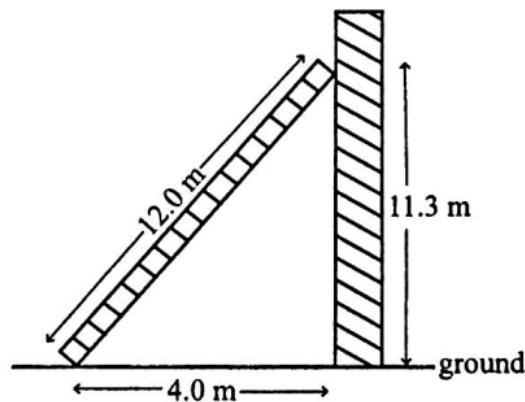


Fig.3.1

A man of mass 60 kg is standing on the ladder's midpoint.

Determine the magnitude of the frictional force that prevents slipping of the ladder.

Q8:NOV 2018 P2

- 2 (a) Define *torque* of a couple.

- 2 (b) Fig. 2.1 shows a wheel supported by a pin P at its centre of gravity.

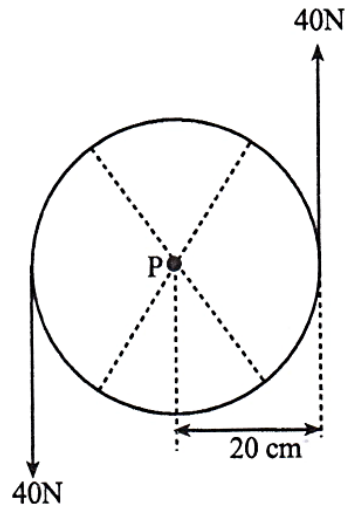


Fig. 2.1

Two parallel forces each 40 N act on the edge of the wheel in the vertical directions. Friction is assumed to be negligible.

- (i) Indicate on Fig.2.1 any **two** other forces present.
(ii) Calculate the torque of the couple on the wheel.

Torque = _____

- (iii) State and explain whether or not the wheel is in equilibrium.

Q9:JUN 2019 P2

- 1 (a) (i) Define *moment of a force*.

- (ii) Explain why the unit of torque is not expressed in Joules.

[2]

- (b) Fig. 1.1 shows a winch used to extract water from a well.

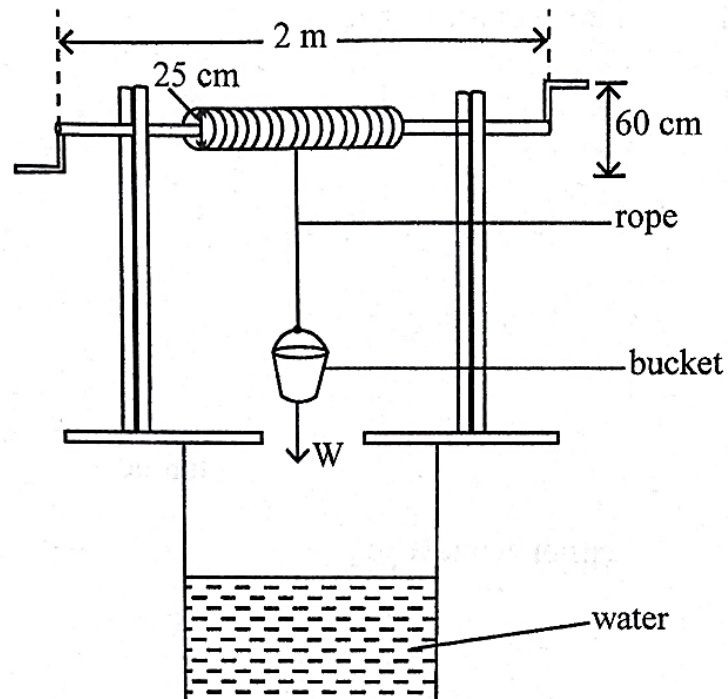


Fig. 1.1

Forces of 40 N are applied normal at the ends of the winch to allow a turning effect. A rope is attached at the end of the winch drum with a bucket full of water. The system is in static equilibrium.

(i) Calculate the

1. torque of the couple,

torque = _____

2. tension in the rope.

tension = _____

(ii) Deduce the weight of the bucket and water.

[4]

Q10:JUN 2007 P3

- 1 (a) (i) Describe an experiment to measure the terminal speed of a solid object through a liquid.
Show how the speed can be determined.
- (ii) If the experiment is repeated without changing any parameter, there is always a marked difference in velocity.
Suggest an explanation for this observation. [9]

- (b) Fig. 1.1 shows a balloon that is attached to the bottom of a pool. The radius of the balloon is 7.8 cm and the mass is 1.23 kg. The density of water is $1\,000\text{ kgm}^{-3}$.

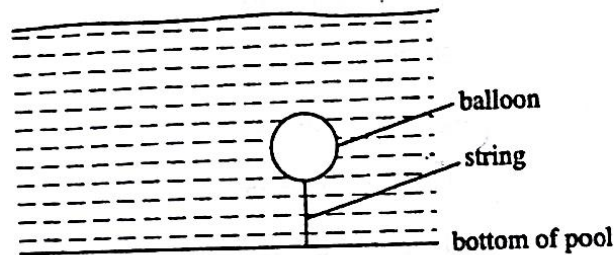


Fig. 1.1

- (i) Sketch a diagram to show the forces acting on the balloon.
- (ii) Calculate the value of the tension in the string.
- (iii) The string is cut and the balloon begins to move. Calculate the acceleration of the balloon as soon as the string is cut.
- (iv) Explain qualitatively whether this value of acceleration can be used for the different parts of the motion.
- (v) State and explain one change that is most likely to take place on the balloon when it comes to rest at the surface. [11]

2.3 FORCES P2&P3 QUESTIONS

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Q11:NOV 2008 P3

- 3 (a) State the conditions for a body to be in equilibrium. [2]
- (b) A Physics teacher used a uniform metre rule to illustrate a point on the board. The rule was held at 32° to the horizontal as shown in Fig. 3.1.

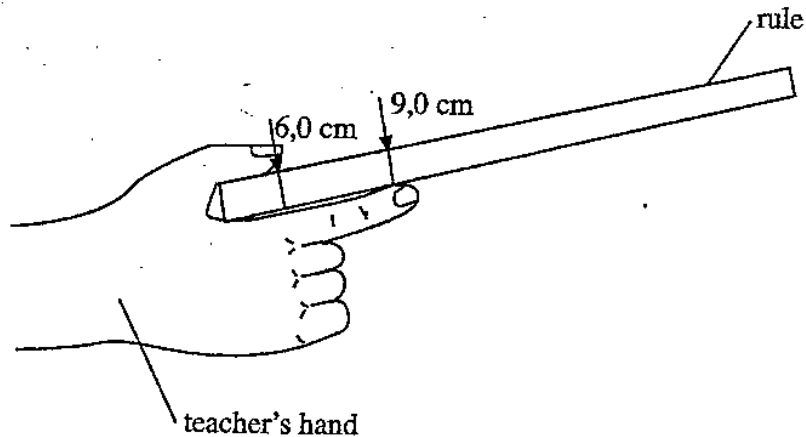


Fig. 3.1

- (i) Make a sketch of Fig. 3.1 and draw all the forces acting on the rule.
- (ii) Given that the mass of the metre rule is 107 g, determine the values of the forces that you have shown in (i). [6]
- (c) State and explain an assumption that you have made in (b)(ii). [2]

Q12:JUN 2011 P3

- 3 (a) Fig. 3.1 shows a steel ball falling freely in a viscous fluid under laminar conditions.

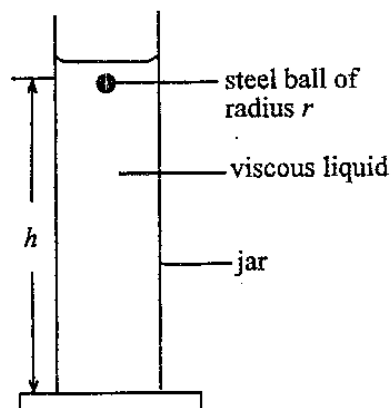


Fig. 3.1

- (i) On a sketch diagram, show all the forces acting on the steel ball.

- (ii) Explain quantitatively how the sphere attains terminal velocity, v_t , giving the expression for v_t .

[7]

- (b) Fig. 3.2 shows a wooden cubic block floating in oil of density 860 kg m^{-3} .

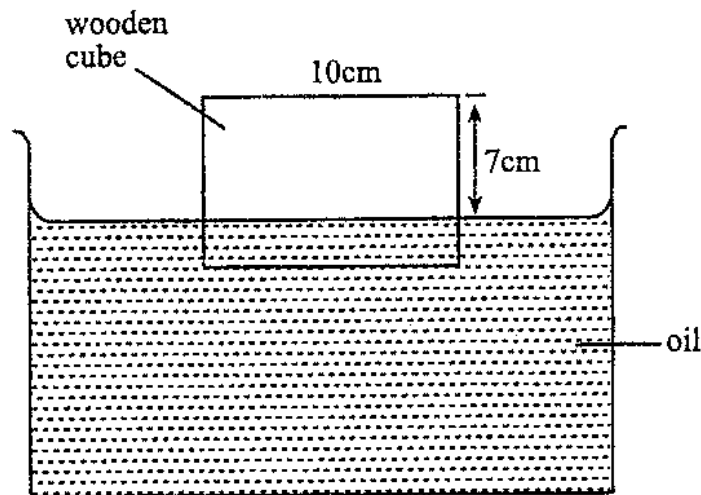


Fig. 3.2

Calculate the weight of the block.

[3]

Q13:JUN 2012 P3

- 2 (ii) A sign post of mass 5.0 kg is hung from the end B of a uniform bar AB, of mass 2.0 kg as shown in Fig.2.1. The bar is hinged to a wall at A and held horizontally by a wire joining B to a point C.

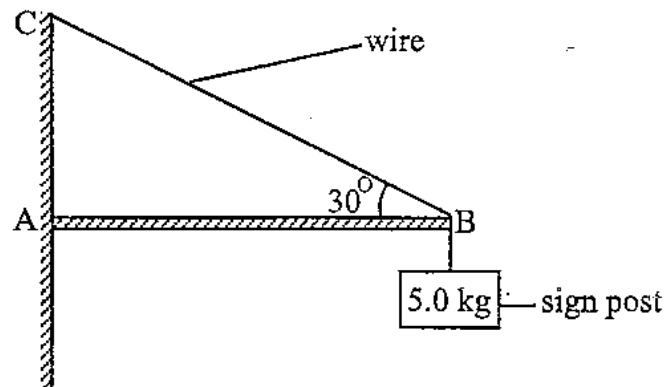


Fig. 2.1

2.3 FORCES P2&P3 QUESTIONS

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Q14:JUN 2014 P3

2 (a) Define the terms *work* and *moment of a force*. [2]

(b) Fig. 2.1 shows a uniform plank of weight 500 N and length 12 m resting on a smooth platform. Halfway up the plank is a bale of cotton of weight 2 500 N.

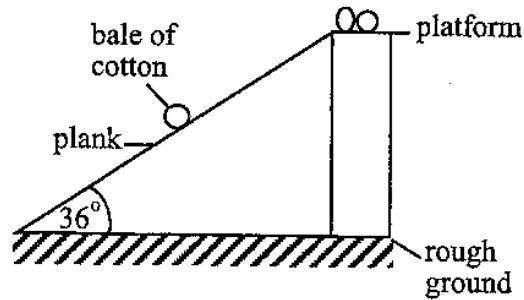


Fig. 2.1

(i) Draw a free-body diagram to show all the forces acting on the plank.

(ii) Calculate the reaction, R , due to the platform on the plank.

[6]

(c) Explain why a worker in construction industry prefers a long crow bar to a short one in moving a large stone from the construction site.

[2]

0

Q15:NOV 2014 P3

2 (a) (i) State two conditions necessary for coplanar forces acting on a body to be in equilibrium.

(ii) Fig. 2.1 shows a 65.0 kg man standing close to the edge of a balcony. The balcony has a weight of 3 450 N acting at a point 2.10 m from the wall. A supporting cable, S, keeps the system horizontal.

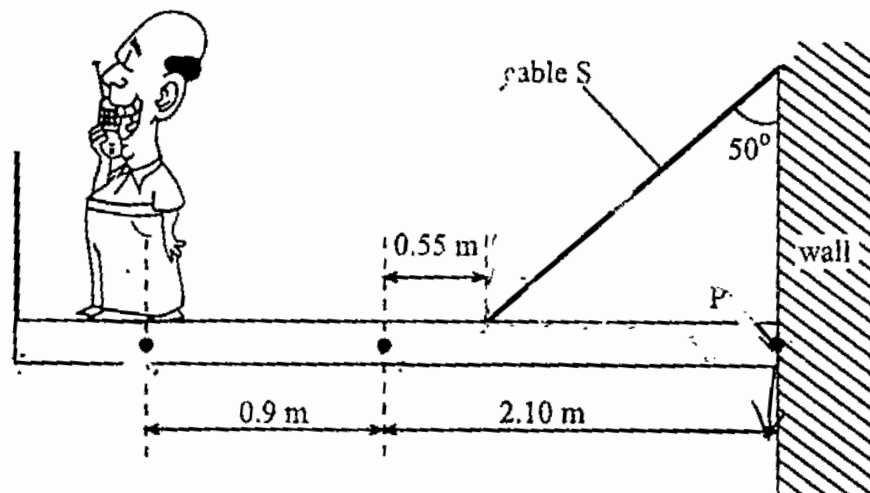


Fig. 2.1

Calculate the

1. tension in S,
2. reaction force at P.

[6]

(b) (i) State Stoke's law.

(ii) Fig. 2.2 shows a moving small steel ball just below the surface of a viscous liquid.

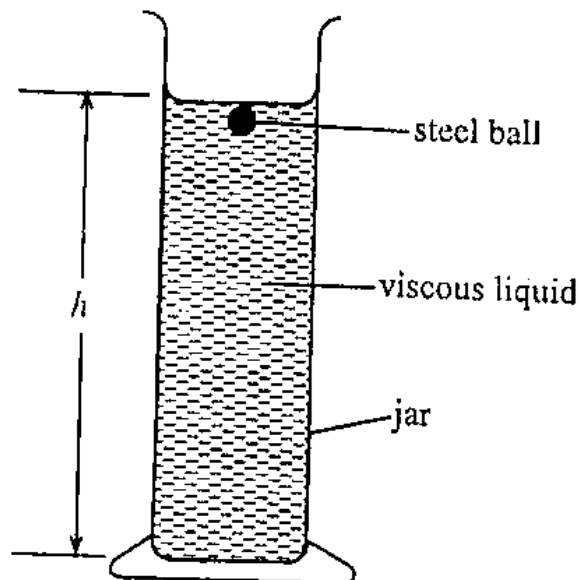


Fig. 2.2

1. Give all the forces acting on the steel ball.
2. The ball falls through the liquid to the bottom. Sketch a graph to show how the height, h , of the ball varies with time given that it attains terminal velocity.

[4]

Q16:NOV 2015 P3

- 3 (a) (i) Define a couple.
- (ii) State two conditions for a body to be in equilibrium.

[3]

- (b) Fig.3.1 shows a horizontal uniform beam of length 8 m and weight 200 N. The beam is pivoted at a wall with one of its ends connected to a metal cable of an angle of depression of 53° . A mass weighing 600 N is placed 2 m from the wall. A reaction force, R , acts at the pivot, P , at an angle, θ , to the horizontal.

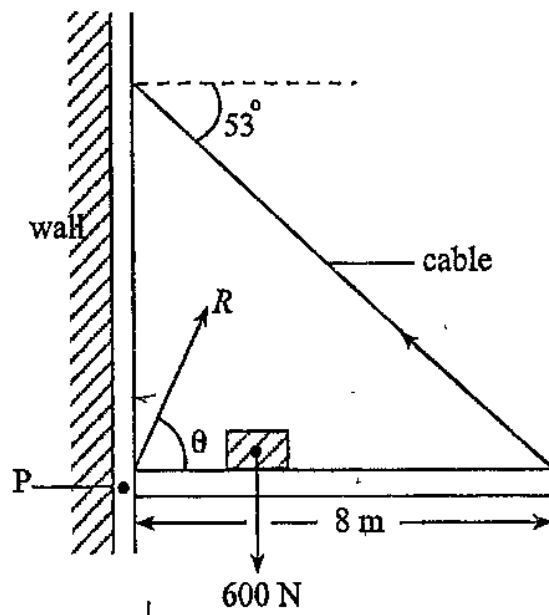


Fig.3.1

Calculate the

- (i) tension in the cable,
- (ii) reaction force, R ,
- (iii) angle θ .

[6]

- (c) Give one example of everyday application of forces in equilibrium.

[1]

Q17:NOV 2016 P3

- 1 (d) Fig. 1.3 shows a model of a front-end loader with a square base of side 1.2 m used to demonstrate the moment of a force. A uniform lever, with first and second arms each of length 1.5 m and weight 20 N, is pinned at the mid-point of the base.

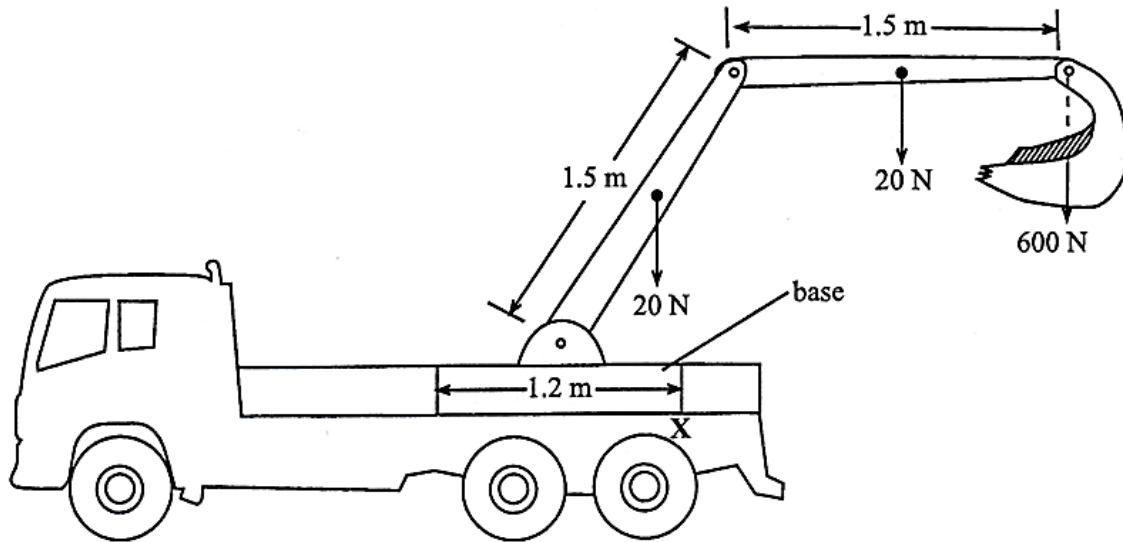


Fig. 1.3

The base is designed such that there is no toppling over at X when the arm is fully extended.

- (i) Explain the term *moment of a force*.
- (ii) Find the minimum weight of the base that just prevents toppling over at point X when the arm is fully extended.

[4]

Q18: NOV 2019 P3

- (e) Fig. 1.1 shows a 10 N uniform rod which is 1.2 m long. The rod is held horizontally by a cable attached at point Y and point Z. The cable makes an angle of 40° with the vertical and a 30 N load is hung at point Y.

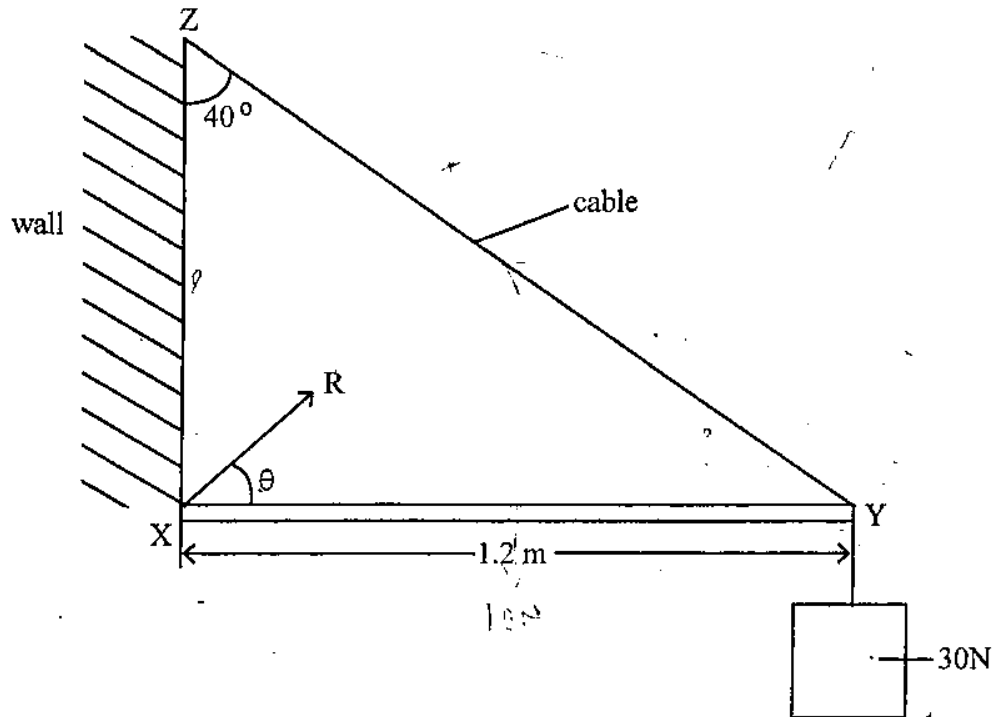


Fig. 1.1

Calculate the

- (i) tension in the cable,
- (ii) angle, θ ,
- (iii) reaction force, R .

[7]

Q19: NOV 2017 P3

- 2 (a) State the conditions for equilibrium. [2]
- (b) Fig. 2.1 shows a tree of length 15 m and mass 300 kg which has fallen over an electrical cable. The tree is suspended 9 m from point A inclined at 60° to the horizontal and its centre of mass is 6 m from point A.

2.3 FORCES P2&P3 QUESTIONS

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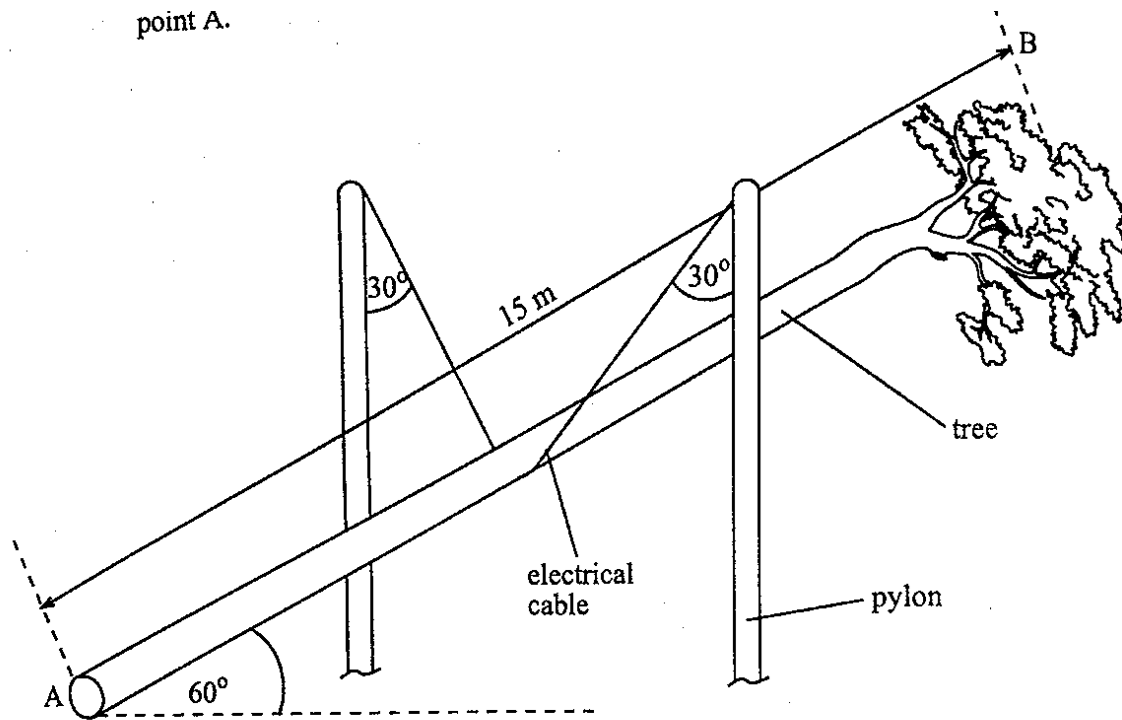


Fig. 2.1

Assuming the pylons remain vertical,

- (i) draw a free body diagram to show all the forces acting in the system,
- (ii) calculate the tension in the cable.
- (iii) Suggest what would happen to the tree if the centre of mass was at the 9 m mark from point A.
- (iv) Given that the cable would support a maximum force of 2 000 N, state with a reason whether or not the cable would break.

[8]